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Theory and Methodology

Preference voting and project ranking using DEA and cross-evaluation

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Abstract

Cook and Kress (1990), using Data Envelopment Analysis (DEA) as their starting point, proposed a procedure to rank order the candidates in a preferential election. Notionally, each candidate is permitted to choose the most favourable weights to be applied to his/her standings (first place, second place, etc. votes) in the usual DEA manner with the additional 'assurance region' restriction that the weight for a j place vote should be more than that for a $j+1$ place vote by some amount. We consider that this freedom to choose weights is essentially illusory when maximum discrimination between the candidates is sought, in which case the weights used to evaluate and rank the candidates are as if imposed externally at the outset. To avoid this, we present an alternative procedure which retains Cook and Kress' central idea but where, as well as using each candidate's rating of him/herself, we now make use of each candidate's ratings of all the candidates. We regard the so-called cross-evaluation matrix as the summary of a self- and peer-rating process in which the candidates seek to interpret the voters' preferences as favourably for themselves, relative to the other candidates, as possible. The problem then becomes one of establishing an overall rating for each candidate from these individual ratings. For this, for each candidate, we use a weighted average of all the candidates ratings of that candidate, where the weights themselves are in proportion to each candidate's overall rating. The overall ratings are therefore proportional to the components of the principal (left-hand) eigenvector of the cross-evaluation matrix. These ideas are then applied to the selection of R&D projects to comprise an R&D program, thus indicating their wider applicability.

Keywords: Data envelopment analysis; DEA; Cross-evaluation; Cross-efficiency; Ranking; Voting

1. Introduction

Cook and Kress (1990) (CK) present an approach to the problem of rank ordering the candidates in a preferential election. In such an election, each voter selects a subset of the candidates and places them in rank order; the poll organizer then establishes for each candidate a *standing* of the number of first, second, third place votes etc. received. The problem is then:

> How should the number of first, second, third place votes etc. that a candidate receives be aggregated into a composite score from which the candidate's position in an overall rank ordering be determined?

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Clearly, one method for resolving this problem is for the poll organiser to impose a predetermined set of weights on each candidate's standing. Thus the composite score, Z_i, of candidate i would be given by:

$$Z_i = \sum_{j=1}^{k} w_j v_{ij}, \tag{1}$$

where v_{ij} is the number of j place votes received by candidate i, w_j is the weight to be given to a j place vote and each voter rank orders k of m, say, candidates. For example, Roberts (1976), in discussing the work of Borda (1781) and Kendall (1962), considers the use of $w_j = m - j$, a version of the well-known Borda–Kendall count.

Cook and Kress consider that schemes involving an imposed set of weights do not provide a "*fair overall* assessment" of a candidate. Their suggestion is to extend the fairness implicit in Data Envelopment Analysis (DEA) (Charnes et al., 1978 (CCR)) where a decision making unit notionally evaluates the efficiency of its performance by choosing the weights to be applied to its outputs and inputs, to the evaluation of the *desirability* of the candidates in an election.

In Section 2 we examine CK's suggestions in detail. We find that the objective of "...provid[ing] the *fairest* possible treatment for each candidate..." is compromised by a second objective of discriminating between the candidates. Such discrimination is achieved by what amounts to an imposed weighting scheme.

In order to retain fairness in evaluation yet achieve discrimination, we suggest a modification to CK's approach in Section 3. Our approach is in the spirit of CK's initial idea which is now combined with the idea of *cross-evaluation* in DEA as proposed by Sexton et al. (1986) and Oral et al. (1991). Alternative forms of cross-evaluation are then discussed in Section 4.

CK point out that a number of problems have a structure similar to that of preference voting, in particular ranking projects. In Section 5 we therefore go on to apply our approach to the problem of evaluating research projects and selecting a portfolio of projects to make up a research program. For this, we use the data in Oral et al. (1991) whose project evaluation model is itself based on cross-evaluation. In so doing, we compare our proposed method with Oral et al.'s.

2. Cook and Kress' approach and its implications

Instead of applying the same externally imposed weighting scheme to all candidates, CK suggest that each candidate be allowed to propose his/her own weights in order to maximize his/her own desirability subject to certain reasonable constraints on the desirabilities of all the candidates, under these weights, and on the weights themselves. CK's *desirability frontier*, consisting of candidates who manage to achieve a desirability of 1, is analogous to the *efficiency frontier* for decision making units in DEA. For candidate i of m candidates CK's model is:

$$Z_{ii}(\epsilon) = \text{Maximize} \sum_{j=1}^{k} w_{ij} v_{ij}, \tag{2a}$$

subject to:

$$Z_{iq}(\epsilon) = \sum_{j=1}^{k} w_{ij} v_{qj} \le 1 \quad \text{for } q = 1,2,\ldots,m \tag{2b}$$

and

$$\begin{cases} w_{ij} - w_{i,j+1} \ge d(j,\epsilon) & \text{for } j = 1,2,\ldots,k-1, \\ w_{ik} \ge d(k,\epsilon), \\ d(.,\epsilon),\ \epsilon \ge 0,\ d(.,0) = 0, \\ d(.,\epsilon) \text{ monotonic increasing in } \epsilon. \end{cases} \tag{2c}$$

Here, w_{ij} is the weight which candidate i places on j place votes; the notation Z_{ii} has been used in the objective function (2a) to emphasize that this is candidate i's own evaluation of his/her own desirability.

Constraints (2b) are the usual DEA constraints i.e. that no candidate q should have a desirability greater than 1 under i's weights. Sexton et al. (1986) refer to Z_{iq} as *cross-efficiency*, which can be thought of here as candidate i's evaluation of candidate q's desirability. We shall return to this concept later.

Constraints (2c) specify a set of plausible conditions on the weights themselves. In principle, they induce an *assurance region* for the weights, as introduced by Thompson et al. (1986). See also Charnes et al. (1990) and Roll and Golany (1993) for a fuller discussion of this extension to the CCR DEA model. The purpose of $d(j,\epsilon)$, which CK refer to as the *discrimination intensity function*, is to ensure that first place votes are valued at least as highly as second place votes which are valued at least as highly as third place votes etc. Thus, when $d(j,\epsilon)=0$, there is a weak ordering of weights

$$w_{i1} \geq w_{i2} \geq \cdots \geq w_{ik}$$

and when $d(j,\epsilon)>0$, there is a strong ordering

$$w_{i1} > w_{i2} > \cdots > w_{ik}.$$

The actual differentials between weights depend on the precise form of $d(j,\epsilon)$ used and, of course, the value selected for ϵ.

Obviously, the desirability scores, Z_{ii}, obtained via (2a–c), and consequently the rank ordering of the candidates, depend on $d(j,\epsilon)$ and the value of the *discriminating factor*, ϵ. A poll organiser planning to use CK's approach, therefore, has two problems: the choice of form for $d(j,\epsilon)$ and the value of ϵ. We will return to the problem of choosing $d(j,\epsilon)$ after first illustrating model 2(a–c) for $d(j,\epsilon)=\epsilon$ on the small problem used by CK, which we reproduce as Table 1.

If we solve (2a–c) with $d(j,\epsilon)=\epsilon$ for the example in Table 1, we find that for:

$$0.000000 \leq \epsilon \leq 0.010753, \quad Z_{bb}=Z_{cc}=Z_{dd}=1;$$
$$0.010753 < \epsilon \leq 0.020408, \quad Z_{bb}=Z_{dd}=1;$$
$$0.020408 < \epsilon \leq 0.023256, \quad Z_{bb}=1.$$

In the first case there is a three-way tie for first place between candidates b, c and d whereas in the last case there is a unique winner, b. As ϵ is increased from 0, (2a–c) becomes more discriminating in the sense that fewer candidates are able to find weights that situate them on the desirability frontier. Further, if $\epsilon > 0.023256$, in this example, there is no feasible solution when (2a–c) is solved on behalf of any of the candidates i.e. the desirability frontier is essentially undefined. CK use this idea to resolve the problem of selecting a value for ϵ, although other problems ensue, as we shall see. On the assumption that $d(j,\epsilon)>0$ for all j, they replace (2a–c) by:

Table 1
Votes achieved by candidates a–f

Candidate	Standing			
	1	2	3	4
a	3	3	4	3
b	4	5	5	2
c	6	2	3	2
d	6	2	2	6
e	0	4	3	4
f	1	4	3	3

Source: Cook and Kress (1990)

$$\text{Maximize } \epsilon \tag{3a}$$

subject to:

$$\sum_{j=1}^{k} w_{?j} v_{qj} \leq 1 \quad \text{for } q=1,2,\ldots,m \tag{3b}$$

and

$$\begin{cases} w_{?j} - w_{?,j+1} = d(j,\epsilon) & \text{for } j=1,2,\ldots,k-1, \\ w_{?k} = d(k,\epsilon). \end{cases} \tag{3c}$$

(Constraints (3c) have been written as equalities rather than "$\geq$" as in (2c) since CK prove that these constraints are satisfied as equalities at optimality in (3a–c).)

The candidate(s) to be placed first in the rank order are signified by the set, P say, of the index(es) of the binding constraint(s) in (3b). We use the notation $w_{?.}$ to make it clear that until (3a–c) has been solved, the identity of the winner(s) is not known. Subsequently we could refer to the weights $w_{?.}$ as $w_{P.}$ The second placed candidate(s) can then be found by re-solving (3a–c) after deleting the constraint(s) indexed by P from (3b). This process could be carried out up to $m-1$ times to order m candidates.

CK concentrate on discrimination intensity functions which can be written as $g(j)h(\epsilon)$ with $h(\epsilon)$ monotonic increasing in ϵ, therefore in what follows here so do we. Any specific function $g(j)$ can be evaluated for $j=1,2,\ldots,k$ to give values

$g_1, g_2, \ldots, g_k$ say, and the equality constraints (3c) allow us to evaluate the weights to a multiplier of h(ϵ) as:

$$w_{?k} = g_k h(\epsilon),$$
$$w_{?k-1} = (g_{k-1} + g_k) h(\epsilon),$$
$$\vdots$$
$$w_{?1} = (g_1 + g_2 + \cdots + g_k) h(\epsilon).$$

These values can be substituted into constraints (3b) and given that the v_{qj} are known data in any particular case, these can be written as:

$$\pi_q h(\epsilon) \leq 1 \quad \text{for } q = 1,2,\ldots,m,$$

where

$$\pi_q = \sum_{j=1}^{k} \sum_{x=j}^{k} g(x) v_{qj}.$$

We can now write (3a–c) as:

$$\text{Maximize } \epsilon \tag{4a}$$

subject to:

$$h(\epsilon) \leq 1/\pi_q \quad \text{for } q = 1,2,\ldots,m \tag{4b}$$

and, by the monotonicity of $h(\epsilon)$, it is easy to see that the winner(s) P is given by the index(es) of the constraint(s) with the biggest π_q. Thus the "delete constraint(s) and re-solve" procedure to establish the rank order is unnecessary in that the candidates would emerge from this in exactly the order of decreasing π_q i.e. the candidates would be ranked in order of decreasing π_q.

We see that $h(\epsilon)$ plays no part in determining the order of the candidates. This is entirely as determined by the function $g(j)$ selected by the poll organiser. In effect candidates are ordered by (1) with π_i playing the role of Z_i. i.e.:

$$Z_i = \pi_i = \sum_{j=1}^{k} w_j v_{ij} \quad \text{with } w_j = \sum_{x=j}^{k} g(x). \tag{5}$$

The above analysis is provided in a slightly different form by CK. The principal conclusion from their analysis is that a discrimination intensity function of the form $d(j,\epsilon) = \epsilon$ in (3a–c) is equivalent to the Borda–Kendall count. Here, however, we conclude that such an equivalence may run counter to the objective of providing the fairest possible treatment of each candidate.

We illustrate the consequences of this analysis using $d(j,\epsilon) = \epsilon/j$ on the example in Table 1. Here $g(j) = 1/j$ and $h(\epsilon) = \epsilon$ so:

$$g_1 = 1,\ g_2 = \tfrac{1}{2},\ g_3 = \tfrac{1}{3},\ g_4 = \tfrac{1}{4}$$

and

$$w_{?1} = \tfrac{25}{12}\epsilon,\ w_{?2} = \tfrac{13}{12}\epsilon,\ w_{?3} = \tfrac{7}{12}\epsilon,\ w_{?4} = \tfrac{3}{12}\epsilon.$$

Applying these weights to the v_{qj} in Table 1 we can write

Maximize ϵ

subject to

$(151/12)\epsilon \leq 1$ Candidate a,
$(206/12)\epsilon \leq 1$ Candidate b,
$(203/12)\epsilon \leq 1$ Candidate c,
$(208/12)\epsilon \leq 1$ Candidate d,
$(85/12)\epsilon \leq 1$ Candidate e,
$(107/12)\epsilon \leq 1$ Candidate f.

It can immediately be seen that the order of the candidates will be d, b, c, a, f, e and $\epsilon_{\max} = 12/208$. The desirabilities of the candidates according to (5) range from 208/12 for d down to 85/12 for e. Essentially, the role played by $\epsilon_{\max}$ ($h(\epsilon_{\max})$ in general) is to scale the weights so that winner's desirability is 1. Thus, having identified the winner as candidate d and evaluated $\epsilon_{\max}$, we can determine:

$$w_{d1} = \tfrac{25}{208},\ w_{d2} = \tfrac{13}{208},\ w_{d3} = \tfrac{7}{208},\ w_{d4} = \tfrac{3}{208},$$

and the candidates' desirabilities now range from 1.0 for d to 107/208 for e.

In addition to the forms of $d(j,\epsilon)$ already mentioned, namely ϵ and ϵ/j, CK also investigate how a function of the form of $d(j,\epsilon) = \epsilon/j!$ in (3a–c) would determine the winner for the example in Table 1. We extend their comparison by including the case where $d(j,\epsilon) = 0$, in (2a–c) and by considering the ordering of all the candidates, not just the winner. The results are summarised in Table 2.

Column 1 of Table 2 shows how DEA without constraints (2c) would order the candidates. There is a four-way tie for first place. Such a lack of

Table 2
Orderings of the 6 candidates a–f according to DEA model (2a–b), DEA model (2a–c) with $d(j,\epsilon) = 0$ and model (3a–c) with various forms of $d(j,\epsilon)$

Position	DEA model (2a–b)	DEA model (2a–c) $d(j,\epsilon) = 0$	model (3a–c) $d(j,\epsilon) =$		
			$\epsilon/j!$	ϵ/j	ϵ
1	b,c,d,e	b,c,d	c	d	b
2			d	b	d
3			b	c	c
4		a	a	a	a
5	a	e,f	f	f	f
6	f		e	e	e
	1	2	3	4	5

discrimination is not unusual in DEA: for example, the five-way tie (out of six candidates) for the site of a high-energy physics laboratory, prior to the introduction of an assurance region, as discussed in Thompson et al. (1986).

The imposition of constraints (2c) with $d(j,\epsilon) = 0$ reduces our four-way tie to a three-way tie (column 2). The effect on the ordering for the various stronger forms of discrimination intensity function within model (3a–c) is shown in columns 3–5. It is clear that all the forms of discrimination agree that all of candidates b, c and d should be preferred to all of candidates a, e and f. However, there is no agreement on who should be judged winner or indeed second or third. Which discrimination intensity function should a poll organiser use?

We conclude this section by reiterating that the starting point of fairness in evaluation, inherent in (2a–c), is possibly compromised by a commitment to discrimination via (3a–c). Specifically, such discrimination is achieved by what amounts to an imposed weighting scheme in which candidates definitely do *not* choose their *own* weights. The precise form that such an imposed weighting scheme should take returns as an open question.

3. Using cross-evaluation

A major appeal of DEA is the flexibility it extends to decision making units, candidates or whatever in evaluating their own performance. Assurance region constraints obviously reduce this flexibility but essentially extend the usefulness of the DEA approach in applications where an unfettered choice of weights would not be theoretically or intuitively appropriate or would be unhelpful in practice.

In the present case, some additional constraints are clearly justifiable. One would imagine that any reasonable person – voter, candidate or poll organiser – would agree that first place votes should weigh at least as much as second place votes, and so on. At the very least the weights should be weakly ordered by including constraints (2c) with $d(j,\epsilon) = 0$. In passing, we note that it is not necessary to explicitly represent the assurance region in this situation. The effect of a weak ordering of the weights can be obtained merely by a simple transformation of the data, v_{qj}, as shown by Golany (1988). For this we replace standing v_{qj} by a *cumulative standing*

$$V_{qj} = \sum_{x=1}^{j} v_{qx}$$

and (2a–c) becomes:

$$Z_{ii} = \text{Maximize} \sum_{j=1}^{k} W_{ij} V_{ij}, \qquad (2a')$$

subject to:

$$Z_{iq} = \sum_{j=1}^{k} W_{ij} V_{qj} \leq 1 \quad \text{for } q = 1,2,\ldots,m. \qquad (2b')$$

The relationship between weights $w_{.}$ and $W_{.}$ is now given by

$$w_{ij} = \sum_{x=j}^{k} W_{ix}.$$

For a full treatment of both weak and strict ordinal relations on weights in DEA see Ali et al. (1991).

If, in some specific election, model (2a′–b′) were to lead to a strong ordering of the candidates, or at least a clear winner, then it would presumably have served well.

Since this situation is perhaps unlikely in practice, there is a need for some form of discrimination. However, achieving discrimination by insist-

ing that $d(j,\epsilon) > 0$ is problematic for the reasons already explicated. Firstly, there is a problem in the choice of the precise form of $d(j,\epsilon)$ to be used. CK investigated three different functions and plausible as they all may be, there is no obvious way of selecting from amongst them, or the infinity of alternatives. Secondly, there is a problem in the choice of the value of ϵ. This is resolved by CK's idea of maximising the discrimination intensity factor in (3a–c). But this procedure leads straight to the third problem in that it corresponds to imposing a weighting scheme specified at the outset, as we have shown. The candidates' "freedom" to choose weights in these circumstances is essentially illusory.

Before going on to propose an alternative approach, it is instructive to reinterpret CK's procedure. Whilst candidates in general do not choose their own weights in (3a–c), it may appear that the *winner* does choose his/her own weights. The winner in (3a–c) effectively does so by establishing a desirability of 1.0 with the weights:

$$w_{Pj} = h(\epsilon_{\max})\left(\sum_{x=j}^{k} g(x)\right) \quad \text{where}$$

$$h(\epsilon_{\max}) = 1/\pi_P.$$

The other candidates are then ranked using these weights. CK's procedure, in a sense, yields the winner and the winner's weights are used to rank order the other candidates. Thus, candidates are ranked according to how they are *cross-evaluated* (Sexton et al., 1986; Oral et al., 1991) by the *winner(s)*, P.

In order to retain the essence of CK's approach, i.e. (2a′b′), but still discriminate between candidates, we are now motivated to investigate how the idea of cross-evaluation, by *all* candidates, within model (2a′–b′) can be used to arrive at an overall rating of each candidate.

When model (2a′–b′) is solved for candidate i, as well as obtaining Z_{ii}, we are also provided with values Z_{iq} which can be thought of as evaluations of q's desirability from i's point of view – within this modelling framework. The values obtained in a complete run of the model can be organised in a matrix $\boldsymbol{Z}$ in which the values across a row i ($Z_{i.}$) represent how candidate i *rates* all candidates and values down column i ($Z_{.i}$) represent how candidate i *is rated by* all candidates. Thus, this matrix can be regarded as the summary of a self- and peer-rating process in which on-diagonal elements represent self-ratings and off-diagonal elements represent peer-ratings. Such a matrix, for the example in Table 1, is shown in Table 3.

Table 3
A cross-evaluation matrix for the example of Table 2

	a	b	c	d	e	f
a	0.813	1.0	0.813	1.0	0.688	0.688
b	0.667	1.0	0.786	0.714	0.35	0.475
c	0.5	0.667	1.0	0.95	0.0	0.167
d	0.5	0.667	0.813	1.0	0.0	0.167
e	0.813	1.0	0.813	1.0	0.688	0.688
f	0.813	1.0	0.813	1.0	0.688	0.688

The problem now is to arrive at an overall rating for each candidate, consistent with the self- and peer-ratings in $\boldsymbol{Z}$, which can then be used to rank the candidates. Our first inclination is to follow Sexton et al. (1986) and regard the column averages of $\boldsymbol{Z}$ as suitable overall ratings. In essence, each candidate is being allowed equal right to interpret the voter's preferences, (i.e. all the candidates' standings) as manifested in $\boldsymbol{Z}$, rather than just the winner(s) according to CK's approach. Thus, each candidate is being accorded a weight of $1/m$ in determining any candidate's overall rating. Applying this idea to the cross-evaluation matrix in Table 3, the column averages and corresponding ranking is as follows:

$$\mathrm{d}(0.944) > \mathrm{b}(0.889) > \mathrm{c}(0.840) > \mathrm{a}(0.684) > \mathrm{f}(0.479) > \mathrm{e}(0.402).$$

However, consider the situation if there were two extra candidates, g and h say, each receiving one third place vote [1]. The cross-evaluation matrix for this situation is given in Table 4. (To reflect the fact that the electorate is finite, with twenty voters, we have deducted one third place vote from each of candidates e and f.)

[1] The authors would like to thank an anonymous referee for providing this example.

Table 4
A cross-evaluation matrix for the example of Table 2 after appending two extra candidates, g and h, who each receive one third-place vote. The standings of candidates e and f have each been reduced by one third-place vote to reflect the finite electorate of twenty voters

	a	b	c	d	e	f	g and h
a	0.813	1.0	0.813	1.0	0.688	0.688	0.063
b	0.667	1.0	0.786	0.714	0.35	0.475	0.0
c	0.5	0.667	1.0	0.95	0.0	0.167	0.0
d	0.5	0.667	0.813	1.0	0.0	0.167	0.0
e	0.813	1.0	0.813	1.0	0.688	0.688	0.063
f	0.813	1.0	0.813	1.0	0.688	0.688	0.063
g and h	0.714	1.0	0.786	0.714	0.429	0.5	0.071

It can be seen that the order of the candidates is now:

$$\mathrm{b}(0.917) > \mathrm{d}(0.887) > \mathrm{c}(0.826) > \mathrm{a}(0.692) > \mathrm{f}(0.484) > \mathrm{e}(0.409) > \mathrm{g} = \mathrm{h}(0.041);$$

there has been a reversal in the positions of candidates b and d consequent on the introduction of the two lowly rated candidates g and h. The 'principle of the independence of irrelevant alternatives' (e.g. Arrow, 1951) has been contravened. However, French (1986) suggests, echoing many other commentators, that this principle is "...arguably the most controversial assumption within social choice theory, not to say within decision theory...".

For our own part, we must be concerned by a rank reversal provoked by a very lowly rated candidate, such as in the example above, but this concern would not necessarily extend to all contexts. In the turbulence following in the wake of the exit from the scene (or entry to the scene) of a *highly* rated candidate, we perhaps should not be surprised to see some change in the order of the candidates.

In order to mitigate the effect observed above, we may relax the assumption that each candidate be accorded a weight of 1/m in the establishment of the overall ratings. Instead, we suggest that each candidate applies a weight in proportion to his/her overall rating rather than uniformly 1/m i.e. a form of "weighted voting" (Tideman, 1976). Thus, if $\boldsymbol{\Theta}$ is the row vector of final ratings we now solve:

$$\left(1/\sum_{i=1}^{m}\Theta_i\right)\boldsymbol{\Theta}\mathbf{Z} = \boldsymbol{\Theta} \tag{6}$$

to obtain candidate i's overall rating, Θ_i.

For the cross-evaluation matrices of Tables 3 and 4 we now obtain:

$$\mathrm{d}(0.928) > \mathrm{b}(0.852) > \mathrm{c}(0.847) > \mathrm{a}(0.643) > \mathrm{f}(0.411) > \mathrm{e}(0.310)$$

and

$$\mathrm{d}(0.926) > \mathrm{b}(0.854) > \mathrm{c}(0.846) > \mathrm{a}(0.644) > \mathrm{f}(0.412) > \mathrm{e}(0.312) > \mathrm{g} = \mathrm{h}(0.022)$$

respectively. The previous rank reversal does not now occur and, perhaps more important, the overall ratings are scarcely changed between the two situations.

In order to solve (6), we proceed iteratively as follows:

$$\begin{aligned} &\text{Step 1.} \quad \boldsymbol{\Theta}^{\mathrm{new}} = \left(1/\sum_{i=1}^{m}\Theta_i^{\mathrm{old}}\right)\boldsymbol{\Theta}^{\mathrm{old}}\mathbf{Z}; \\ &\text{Step 2.} \quad \boldsymbol{\Theta}^{\mathrm{old}} = \boldsymbol{\Theta}^{\mathrm{new}}. \end{aligned} \tag{7}$$

Initially we set $\boldsymbol{\Theta}^{\mathrm{old}}$ as $(1,1,\ldots,1)$ so $1/\Sigma\Theta_i^{\mathrm{old}}$ is $1/m$ and the first iteration gives the vector of simple column averages of $\mathbf{Z}$. This is then refined in subsequent iterations. The algorithm in (7) is essentially the "Power Method" (Atkinson, 1978) for the principal eigenvalue/eigenvector of $\mathbf{Z}$. Whilst this method is not particularly efficient as a means of obtaining principal eigenvalue/eigenvectors of arbitrary matrices, it works very well for cross-evaluation matrices, converging to 3 decimal places in about five iterations.

The resemblance of (7) to the Power Method is not surprising, of course, since (6) can be rewritten as:

$$\boldsymbol{\Theta}\mathbf{Z} = \left(\sum_{i=1}^{m}\Theta_i\right)\boldsymbol{\Theta},$$

whereupon $\boldsymbol{\Theta}$ can be seen as the left-hand eigenvector of $\mathbf{Z}$ scaled to an eigenvalue of $\Sigma\Theta_i$. Our proposal can now be seen to be reminiscent of Wei's rating method (Wei, 1952; but see Cook and Kress, 1992) and also Saaty's "Analytic Hier-

archy Process" (Saaty, 1977, 1994) which both use the principal eigenvector of a matrix as representing an overall rating. Of course, the meanings of the matrices and the manner of their determination in these two cases are somewhat different to our cross-evaluation matrix.

4. Alternative forms of cross-evaluation

Actually, determining $\boldsymbol{Z}$ via model (2a′–b′) is not quite as unproblematic as we have hitherto suggested, but the resolution of the problem fits in well with what a collection of self-interested individuals might be expected to do. The problem arises because, although the leading diagonal elements in $\boldsymbol{Z}$, Z_{ii}, are uniquely determined, the off-diagonal elements are not necessarily so, and there is consequently a potential arbitrariness in the rating vector, $\boldsymbol{\Theta}$. For example, candidate b could establish a desirability $Z_{bb} = 1.0$ with the weights $w_{b1} = w_{b2} = w_{b3} = w_{b4} = 0.0625$ or $w_{b1} = w_{b2} = w_{b3} = 0.0714$ and $w_{b4} = 0$ amongst others. The value Z_{bb} is the same in each case but clearly the off-diagonal values $Z_{b.}$ may differ. However, given that a candidate would seek to win the election then it seems natural to suggest that he/she would attempt to minimise the other candidates' desirabilities whilst maximising his/her own i.e. to find the largest possible principal diagonal element and the smallest possible off-diagonal elements in Z. Thus, we combine the suggestions of Sexton et al. (1986) and Oral et al. (1991), and incorporate into the objective function of (2a′–b′) a secondary objective of minimising each of the other candidates' desirabilities in turn. We now write (2a′–b′) as:

$$\text{Maximise } (P_1 Z_{ii} - P_2 Z_{iq}) \quad \text{for all } i,q, \tag{8a}$$

where

$$P_1 \gg P_2$$

subject to:

$$Z_{iq} = \sum_{j=1}^{k} W_{ij} V_{qj} \leq 1 \quad \text{for } q = 1,2,\ldots,m. \tag{8b}$$

We now declare that Tables 3 and 4 were obtained via (8a–b).

Table 5
The benevolent cross-evaluation matrix corresponding to the aggressive version in Table 3

	a	b	c	d	e	f
a	0.813	1.0	0.813	1.0	0.688	0.688
b	0.813	1.0	1.0	1.0	0.688	0.688
c	0.744	1.0	1.0	1.0	0.39	0.5
d	0.813	1.0	1.0	1.0	0.688	0.688
e	0.813	1.0	0.813	1.0	0.688	0.688
f	0.813	1.0	0.813	1.0	0.688	0.688

In contrast to the objective function (8a) which may be termed *aggressive* cross-evaluation, we could also use

$$\text{Maximise } (P_1 Z_{ii} + P_2 Z_{iq}) \text{ for all } i,q, \tag{8a′}$$

which we might term *benevolent* cross-evaluation. Here the maximum possible values for the off-diagonal elements in $\boldsymbol{Z}$ are found. Table 5 reports the benevolent cross-evaluation matrix corresponding to what is now seen as the aggressive cross-evaluation matrix of Table 3. The rank order and overall ratings, Θ_i, of the six candidates given by benevolent cross-evaluation is

$$\text{d} = \text{b}(1.000) > \text{c}(0.922) > \text{a}(0.800) > \text{f}(0.653) > \text{e}(0.633).$$

As suggested by Sexton et al. (1986) and indicated by the two rankings in this example, the aggressive version is likely to provide a sharper discrimination between the candidates than the benevolent.

It should be noted that the terms "aggressive" and "benevolent" as applied to cross-evaluation originate with Sexton et al. (1986). However, their implementation of these ideas is slightly different from those above, which originate in Oral et al. (1991). The objective functions of Sexton et al. (1986) would be

$$\text{Maximise } \left(P_1 Z_{ii} \pm \sum_{q \neq i} P_2 Z_{iq} \right) \quad \text{for all } i, \tag{9}$$

where the "−" in "±" signifies aggression as in (8a) and the "+" benevolence as in (8a′). Only m LPs are needed to be run to fill out the cross-evaluation matrix rather than the $O(m^2)$ implicit in (8a) and (8a′). However, we prefer the latter

versions since the resulting cross-evaluation matrices are canonical in the sense that the off-diagonal elements are at their lowest or highest possible values.

5. Selecting projects to form an R&D program

In order to examine our use of cross-evaluation in a more challenging environment, and to extend its range of application beyond preference voting, we turn to the problem of selecting R&D projects to form an R&D program from amongst a set of proposed projects. Oral et al. (1991) develop a DEA-based method for this purpose. As in our proposal above, their method uses the idea of self- and cross-evaluation where the starting point for their selection model is a matrix Z obtained according to our (8a′) above.

Thus each project is placed in the *best* possi-

Table 6
Data on 37 R&D projects relating to their expected performance on five criteria and their cost

R&D project	Indirect economic contribution	Direct economic contribution	Technical contribution	Social contribution	Scientific contribution	Project cost
1	67.53	70.82	62.64	44.91	46.28	84.20
2	58.94	62.86	57.47	42.84	45.64	90.00
3	22.27	9.68	6.73	10.99	5.92	50.20
4	47.32	47.05	21.75	20.82	19.64	67.50
5	48.96	48.48	34.90	32.73	26.21	75.40
6	58.88	77.16	35.42	29.11	26.08	90.00
7	50.10	58.20	36.12	32.46	18.90	87.40
8	47.46	49.54	46.89	24.54	36.35	88.80
9	55.26	61.09	38.93	47.71	29.47	95.90
10	52.40	55.09	53.45	19.52	46.57	77.50
11	55.13	55.54	55.13	23.36	46.31	76.50
12	32.09	34.04	33.57	10.60	29.36	47.50
13	27.49	39.00	34.51	21.25	25.74	58.50
14	77.17	83.35	60.01	41.37	51.91	95.00
15	72.00	68.32	25.84	36.64	25.84	83.80
16	39.74	34.54	38.01	15.79	33.06	35.40
17	38.50	28.65	51.18	59.59	48.82	32.10
18	41.23	47.18	40.01	10.18	38.86	46.70
19	53.02	51.34	42.48	17.42	46.30	78.60
20	19.91	18.98	25.49	8.66	27.04	54.10
21	50.96	53.56	55.47	30.23	54.72	74.40
22	53.36	46.47	49.72	36.53	50.44	82.10
23	61.60	66.59	64.54	39.10	51.12	75.60
24	52.56	55.11	57.58	39.69	56.49	92.30
25	31.22	29.84	33.08	13.27	36.75	68.50
26	54.64	58.05	60.03	31.16	46.71	69.30
27	50.40	53.58	53.06	26.68	48.85	57.10
28	30.76	32.45	36.63	25.45	34.79	80.00
29	48.97	54.97	51.52	23.02	45.75	72.00
30	59.68	63.78	54.80	15.94	44.04	82.90
31	48.28	55.58	53.30	7.61	36.74	44.60
32	39.78	51.69	35.10	5.30	29.57	54.50
33	24.93	29.72	28.72	8.38	23.45	52.70
34	22.32	33.12	18.94	4.03	9.58	28.00
35	48.83	53.41	40.82	10.45	33.72	36.00
36	61.45	70.22	58.26	19.53	49.33	64.10
37	57.78	72.10	43.83	16.14	31.32	66.40

Source: Oral et al. (1991).

ble light as seen by itself and each of the other projects. No specific argument is advanced for preferring a benevolent rather than an aggressive approach in that Oral et al. (1991) do not consider the latter. An argument could be advanced for using an aggressive formulation because the projects are in competition to get into the program. However, since the major interest is in finding a group of projects to make up the program rather than an ordering per se, benevolence, in the eye of some neutral evaluator, may be seen as appropriate in this context.

Having obtained $\boldsymbol{Z}$, they then convert it into a *concordance matrix* $\boldsymbol{C}$ in which C_{ij} is the proportion of the projects which evaluate project i at least as highly as project j. This is done by comparing columns i and j of $\boldsymbol{Z}$ pairwise and counting the number of times $Z_{.i}$ is greater than or equal to $Z_{.j}$. This matrix $\boldsymbol{C}$ then provides, together with the cost of each project, the data for a non-linear, mixed-binary program which is used iteratively to build the research program subject to an overall budget constraint.

They illustrate their method using an example of 37 project proposals relating to the Turkish iron and steel industry. Each project is characterised by five output measures – direct economic contribution, indirect economic contribution, technological contribution, scientific contribution and social contribution – relative to its consumption of resources expressed in monetary terms. We reproduce their data in Table 6.

Instead of Oral et al.'s somewhat complicated processing of $\boldsymbol{Z}$ (Oral et al., 1991) to arrive at a research program we consider a simpler alternative. In a nutshell, our procedure is to rank order the projects by decreasing values of Θ_i, and then accept them into the program sequentially until the budget for the program is exhausted. In more detail:

1. Compute the cross-evaluation matrix $\boldsymbol{Z}$ via (8a′–b).
2. Determine an overall rating, Θ_i, for each project via (7).
3. Sort projects into a list in order of decreasing Θ_i.
4. Sequentially accept projects into the program working down the list until the budget is exhausted.

Table 7
A comparison of the program developed by Oral et al. (1991) with that which would be obtained by our approach

Project	Overall rating Θ_i	Oral et al.'s program?	Green et al.'s program?	Project cost
35	1.000	yes	yes	36.0
17	0.975	yes	yes	32.1
31	0.866	yes	yes	44.6
16	0.780	yes	yes	35.4
36	0.759	yes	yes	64.1
18	0.715	yes	yes	46.7
27	0.712	yes	yes	57.1
34	0.699	yes	yes	28.0
37	0.684	yes	yes	66.4
23	0.655	yes	yes	75.6
26	0.632	yes	yes	69.3
1	0.614	yes	yes	84.2
14	0.611	yes	yes	95.0
32	0.606	no	yes	54.5
21	0.565	yes	yes	74.4
29	0.559	yes	yes	72.0
11	0.544			76.5
30	0.538			82.9
15	0.537			83.8
12	0.530	no	yes	47.5
6	0.528			
10	0.525			
2	0.519			
19	0.484			
24	0.476			
22	0.472			
13	0.466			
5	0.457	yes	no	75.4
4	0.457			
9	0.444			
7	0.436			
8	0.409			
33	0.404			
25	0.359			
28	0.331			
20	0.307			
3	0.259			
Program cost		956.3	982.9	

The result from applying this straightforward algorithm to Oral et al.'s data is shown in Table 7 where it can be seen that of Oral et al.'s suggested program of sixteen projects (Oral et al., 1991), namely 1, 5, 14, 16, 17, 18, 21, 23, 26, 27,

29, 31, 34, 35, 36 and 37, we agree on fifteen. Essentially, we substitute project 32 (overall rating 0.606) for project 5 (0.457) and then we would also include project 12 (0.530) in order to absorb as much of the available budget of 1000.00 monetary units as possible. Projects 11, 30 and 15, whilst having slightly better ratings than 12, cannot be accommodated within the residual budget after choosing the higher rating projects.

Clearly, our result is substantially equivalent to that obtained by Oral et al. (1991). Furthermore, our procedure is somewhat less complex. From the common starting point of the cross-evaluation matrix $\boldsymbol{Z}$, we establish an overall rating for each project Θ_i via three iterations of (7). Our subsequent operations on these ratings are conceptually transparent and use only simple arithmetic. Contrast this with the approach of Oral et al. (1991) where the cross-evaluation matrix $\boldsymbol{Z}$ is firstly converted into a concordance matrix $\boldsymbol{C}$ which is then used to generate an initial non-linear, mixed-binary program with approximately m^2 (37^2 in this case) binary variables and a similar number of constraints – considerably more of each after linearization (Kettani and Oral, 1990). The (linearized) mixed-binary program is then run a number of times (in this case 10) to select the projects to comprise the program. This is not necessarily to criticize Oral et al.'s approach which, indeed, we take as a benchmark for our own. The two approaches are mutually supportive, but one might conclude there is merit in the greater transparency and generality of our approach.

6. Conclusions

Our starting point in this paper is Cook and Kress' DEA-based approach for ordering candidates in a preferential election (Cook and Kress, 1990), i.e. model (2a–c). However, we find the sequel to this, model (3a–c), where maximum discrimination between the candidates is sought, problematic. Whilst maximizing discrimination in the manner suggested seems entirely reasonable, we have seen that it amounts to imposing a single set of weights on all candidates from the outset. Once the discrimination intensity function, $d(j,\epsilon)$, is fixed, the weighting scheme and the resultant ordering may be calculated without resorting to optimization; the panoply of linear programming need not be invoked at all. This is, of course, an advantage to a poll organiser or evaluator firmly committed to some specific $d(j,\epsilon)$. However, the problem of what constitutes an appropriate $d(j,\epsilon)$ remains.

So as to retain the essence of Cook and Kress' intention, we have examined the use of the weakest possible discrimination intensity function, $d(j,\epsilon) = 0$, in (2a–c) which leads us to (8a–b) or (8a′–b). This implements a minimal requirement that a first place vote should count at least as much as a second place vote, and a second place at least as much as a third place, etc. If the resulting Z_{ii} values imply an ordering of the candidates sufficiently discriminated for the object of the exercise, no further analysis is required. If this is not the case, we have considered making use of the previously ignored Z_{iq} values obtained as a by-product of the analysis. We regard the totality of these Z_{ii} and Z_{iq} values as the summary of a collective self- and peer-rating process in which each candidate rates him/herself and all the other candidates.

The problem then becomes one of establishing an overall rating for each candidate. For this purpose we use the principal left-hand eigenvector of the cross-evaluation matrix, the components of which vector happen to be a weighted average of each candidate's ratings *by* all the candidates. The weightings in this average are in fact proportional to each candidate's overall rating. One justification for considering this idea is the recognition that Cook and Kress' procedure can be interpreted such that the order of the candidates is determined by their ratings as seen by the winning candidate. Our proposal extends this right to all the candidates in proportion to their final overall rating.

Furthermore, the close correspondence in the result obtained by Oral et al.'s method for evaluating and selecting research projects and that obtained here suggest that our approach has wider applicability.

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